

Instance Segmentation of Solar Filaments in GONG $H\alpha$ Images with a Calibrated Ensemble of Detectors

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Abstract—Solar filaments are clouds of cool plasma held above the solar surface by magnetic fields, and their eruptions are a frequent source of coronal mass ejections. Catalogs of filaments are built from full-disk $H\alpha$ images, and segmenting each filament as a separate object in these images is difficult because filaments are thin, fade toward their ends, and are often marked differently by different annotators. This paper describes the system we built for the IEEE BigData Cup 2026 Solar Filament Segmentation Challenge, which uses GONG $H\alpha$ images from the MAGFiLO dataset and scores submissions with panoptic quality (PQ). Four instance segmentation detectors and a semantic segmentation network propose filaments, and a crop-level U-Net redraws every proposed mask. Isotonic calibration puts the scores of all models on a common scale. Each final mask is then a weighted vote of the overlapping masks from all models. We describe the design choices, the evaluation procedure we used to accept or reject them, and the results. On held-out validation images, PQ is 0.461 for the full system against 0.450 for the four calibrated detectors alone, and the system scores 0.40 on the public leaderboard. We also report the organizers’ overlap statistics, the coverage of thin filament structures such as barbs, results by observatory, and the agreement between two human annotators on the same images (PQ 0.31–0.34).

Index Terms—solar filaments, H-alpha, GONG, instance segmentation, panoptic quality, ensemble learning

I. INTRODUCTION

Filaments are dense, cool plasma suspended by magnetic fields above polarity inversion lines. In $H\alpha$ images of the full solar disk they appear as dark, elongated structures, and when they erupt they often produce coronal mass ejections and flares. The Global Oscillation Network Group (GONG) [1] records $H\alpha$ images of the full disk around the clock from six observatories. Segmenting filaments automatically in this stream would allow catalogs to be built and individual filaments to be followed over time.

The MAGFiLO dataset [2] provides hand-drawn outlines and spines (center lines) of filaments in GONG images, and the IEEE BigData Cup 2026 Solar Filament Segmentation Challenge [3] uses it to evaluate instance segmentation with panoptic quality (PQ) [4]. Under this metric every filament must be found and outlined as a separate object. Three properties of the data make this hard. Filaments are thin, often only a few pixels wide in a 2048×2048 image, and they fade gradually toward their ends. Neighboring filaments touch or break into pieces. And 42% of the training images were

annotated by two or three people, who often disagree about which faint structures are filaments and where they end.

Earlier work has applied deep networks to filament segmentation in $H\alpha$ images [5], [6], [7], and Diercke et al. [8] combined an object detector with a segmentation network in a semi-supervised setting. On MAGFiLO, Solomon et al. [9] guided the attention of a U-Net with a learned edge map to segment barbs, which matter for determining the chirality of a filament. We look at how well our system covers such thin structures in Section IV-C. Outside solar physics, combining the outputs of several detectors is common in object detection, for example by averaging the boxes of overlapping detections [10]. Our system also detects filaments first and segments them afterwards, but it combines several detectors and a semantic segmentation model, since in our experiments no single model came close to the combination.

Below we describe the final system and the experiments that shaped it, including the many alternatives that did not help, and look at the errors that remain. Section II covers the data and the evaluation procedure, Section III the system, Section IV the results, and Section V practical aspects and limitations.

II. DATA AND EVALUATION

Data. The challenge training set contains 707 GONG $H\alpha$ images of 2048×2048 pixels from six observatories, taken between 2011 and 2022. Because some images were annotated by several people, there are 1,154 annotation sets with 8,199 filament outlines in total, each with a spine. The test set has 180 images, and the public leaderboard is computed on about half of them. We used only the training annotations provided by the challenge and no MAGFiLO annotations of test images.

Validation split. We held out 10% of the images (70 images with 116 annotation sets and 908 filaments). The split is by image, so that an image annotated by two people never appears in both training and validation. Each annotation set is scored on its own, as in the challenge. PQ is the sum of the IoUs of matched pairs divided by $|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|$, where a prediction and an annotation match if their IoU exceeds 0.5, and the counts are pooled over all images, following the organizers’ evaluation notebook. We also report its two factors, segmentation quality (SQ, the mean IoU of matched pairs) and recognition quality ($RQ = PQ/SQ$).

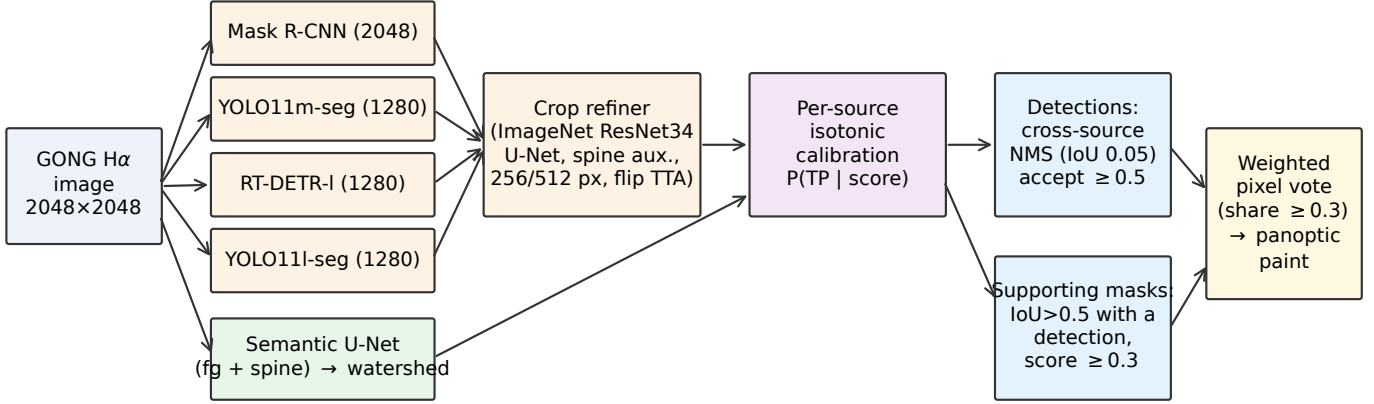


Fig. 1. Overview of the system. Proposals from four detectors are redrawn by a crop refiner, scores are calibrated per model, non-maximum suppression across models selects the detections, and each detection’s final mask is a weighted vote over the overlapping masks from all models. The semantic U-Net contributes both detections and masks.

Evaluation procedure. Some parts of the system, in particular the score calibration, are fitted on validation data, so scores measured on the same data would be optimistic. For each comparison we split the 70 validation images into two halves, fitted the calibration on one half and scored the other. We repeated this for five random assignments of images to halves and used the same five assignments for every configuration. To judge whether a difference between two configurations is real, we used a paired bootstrap over images with 2,000 resamples. The five splits share the same images and the same trained models, so they are not independent. They show how much the result depends on the calibration data, but not how much it would change with a different validation set or retrained models. As a reference, retraining one detector with a different random seed changed PQ by up to 0.005, and we did not treat smaller differences as meaningful. In the later stages of the project we kept a change only if it improved PQ on all five splits.

III. SYSTEM

Fig. 1 gives an overview. The code is available at <https://github.com/biconcavelens/filament-segmentation-2026>. The training settings of all models are listed in Table I.

A. Instance detectors

We use four detectors that differ in architecture and input size: Mask R-CNN [11] (torchvision ResNet-50-FPN v2, pretrained on COCO) at the full 2048-pixel resolution, YOLO11m-seg and YOLO11l-seg [12] at 1280 pixels, and RT-DETR-l [13] at 1280 pixels. In early experiments, many boxes that were placed correctly received very low confidence scores. Increasing the weight of the classification loss reduced this problem (3 times the default for Mask R-CNN, $cls=1.5$ for YOLO and $cls=2.0$ for RT-DETR). For RT-DETR, the median score of boxes overlapping a filament with IoU above 0.5 rose from 0.03 to 0.66. Training Mask R-CNN at the full

TABLE I
TRAINING SETTINGS (EACH MODEL WAS TRAINED ON A SINGLE 16 GB KAGGLE GPU OR AN 8 GB LAPTOP RTX 4060).

Model	Init., input	Training
Mask R-CNN	COCO, 2048	SGD lr 2.5×10^{-3} , 6 epochs, batch 1, cls. loss $\times 3$, flips
YOLO11m-seg	COCO, 1280	40 epochs, batch 4, $cls=1.5$
YOLO11l-seg	COCO, 1280	103 epochs (time budget), $cls=1.5$
RT-DETR-l	COCO, 1280	60 epochs, batch 2, $cls=2.0$
Crop refiner	ImageNet R34, 256 crop	Adam 3×10^{-4} , cosine, 15 epochs, batch 16, rotation, flips, brightness
Semantic U-Net	ImageNet ConvNeXt-T, 1024 crop	AdamW 3×10^{-4} , cosine, 62 epochs (3 h), batch 4, rotations, flips, photometric

image resolution rather than at 1333 pixels improved a two-detector version of the system by 0.0035 PQ.

B. Crop refiner

The masks produced by the detectors’ own mask heads are coarse, so we redraw every proposal with a separate network. A square crop 1.8 times the size of the proposal is cut from the full-resolution image, resized to 256×256 pixels, and passed to a U-Net [14] with an ImageNet-pretrained ResNet-34 encoder [15], [16]. Besides the mask, the network also predicts the annotated spine as a second output, which is used only during training. The loss is BCE plus Dice on the mask (weight 0.7) and BCE on the spine (weight 0.3), and at test time we average the predictions for four flipped versions of the crop. Our first refiner was a U-Net of similar size trained from scratch. Replacing it with the pretrained encoder lowered the validation loss from 0.103 to 0.092 and improved PQ of the four-detector system by 0.006 to 0.012 on all five splits. A second copy of the refiner, trained on 512×512 crops, is used only to provide additional masks for the vote described below.

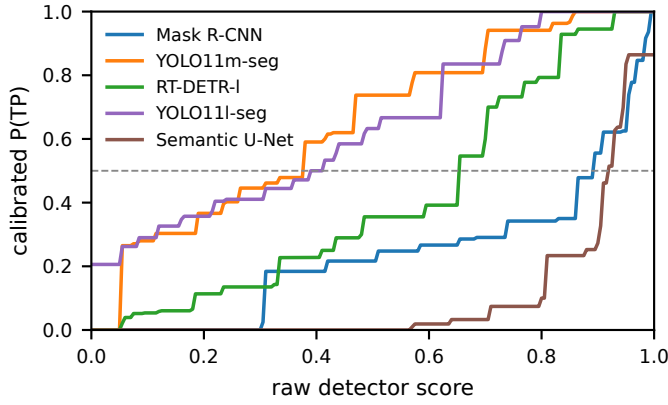


Fig. 2. Isotonic calibration of each model’s score into the probability that a candidate is correct, fitted on validation candidates. The dashed line marks the threshold for accepting a detection.

C. Semantic segmentation model

The fifth model is a U-Net with an ImageNet-pretrained ConvNeXt-Tiny encoder [17] that labels every pixel. It is trained at full resolution on 1024×1024 crops with two outputs, the filament foreground and a 5-pixel-wide drawing of the spines. The foreground is trained with focal loss [18] plus Dice loss [19] and the spine with BCE plus Dice. To turn its output into separate filaments, we threshold the foreground at 0.5 and take connected regions of the spine map above 0.3 as seeds. A watershed grows the seeds into the foreground, which splits touching filaments along their spines. Regions smaller than 100 pixels are removed, and each filament is scored with its mean foreground probability.

D. Combining the models

The scores produced by different models mean different things (Fig. 2). For example, a candidate has a 50% chance of being a correct detection at a score of 0.38 for YOLO11m-seg, 0.65 for RT-DETR and 0.89 for Mask R-CNN. We map each model’s scores to the probability that a candidate is correct, using isotonic regression [20] fitted on validation candidates. A candidate counts as correct if its redrawn mask overlaps an annotated filament with IoU above 0.5.

Algorithm 1 shows how the candidates are combined in each image. Detections are chosen from the candidates of the four detectors and the semantic model. Candidates are taken in order of decreasing calibrated score, and a candidate is accepted if its score is at least 0.5 and it overlaps no accepted detection by more than IoU 0.05. The mask of each accepted detection is then replaced by a weighted vote over all candidates that describe the same filament. Every remaining candidate with a score of at least 0.3 joins the detection it overlaps most, provided that IoU exceeds 0.5. The final mask keeps the pixels covered by at least 30% of the score-weighted masks in the group. This is similar in spirit to weighted box fusion [10], applied to masks. The masks taking part in the vote come from more sources than the detections. Besides the five sources above, they include the masks from the 512×512

Algorithm 1 Selecting detections and voting on their masks (one image)

Require: candidates (m_i, p_i, s_i) with mask, calibrated score and source; set $\mathcal{D}$ of sources allowed to create detections

- 1: $D \leftarrow \emptyset$
- 2: **for** i in order of decreasing p_i **do**
- 3: **if** $s_i \in \mathcal{D}, p_i \geq 0.5, \text{IoU}(m_i, m_j) \leq 0.05$ for all $j \in D$ **then** $D \leftarrow D \cup \{i\}$
- 4: **end if**
- 5: **end for**
- 6: $G_d \leftarrow \{d\}$ for each $d \in D$
- 7: **for** each candidate $j \notin D$ with $p_j \geq 0.3$ **do**
- 8: $d^* \leftarrow \arg \max_{d \in D} \text{IoU}(m_j, m_d)$
- 9: **if** $\text{IoU}(m_j, m_{d^*}) > 0.5$ **then** $G_{d^*} \leftarrow G_{d^*} \cup \{j\}$
- 10: **end if**
- 11: **end for**
- 12: **for** each $d \in D$ **do**
- 13: $\hat{m}_d(x) \leftarrow [\sum_{k \in G_d} p_k m_k(x) / \sum_{k \in G_d} p_k \geq 0.3]$
- 14: **end for**
- 15: draw $\hat{m}_d$ in order of decreasing p_d onto free pixels; drop pieces < 20 px

TABLE II
VALIDATION RESULTS WITH CALIBRATION FITTED ON HELD-OUT HALVES: MEAN PQ (STANDARD DEVIATION) OVER FIVE RANDOM SPLITS, SQ, RQ, AND THE CHANGE IN PQ RELATIVE TO THE FOUR CALIBRATED DETECTORS (LOWEST AND HIGHEST VALUE OVER THE SPLITS, AND THE NUMBER OF SPLITS WHOSE 95% BOOTSTRAP INTERVAL LIES ABOVE ZERO). ROWS STARTING WITH + ADD TO THE FOUR-DETECTOR SYSTEM. LAST COLUMN: PUBLIC LEADERBOARD.

Configuration	PQ	SQ	RQ	Δ PQ range (sig.)	Pub.
Mask R-CNN	.429 (.004)	.682	.629		
YOLO11m-seg	.437 (.002)	.684	.639		
RT-DETR-l	.436 (.002)	.686	.636		
YOLO11l-seg	.420 (.004)	.687	.611		
Semantic U-Net	.396 (.003)	.669	.591		
4 detectors + NMS	.450 (.003)	.682	.660	–	.40
+ semantic, no vote	.448 (.001)	.678	.661	[−.006, +.003]	(0/5)
+ vote, 4 sources	.453 (.003)	.682	.664	[+.003, +.004]	(0/5)
+ vote, 12 sources	.457 (.002)	.684	.668	[+.005, +.009]	(3/5)
+ semantic, 5 sources	.455 (.001)	.680	.668	[+.001, +.009]	(0/5)
+ semantic, 13 sources	.461 (.002)	.683	.675	[+.007, +.016]	(3/5)

refiner and from running the four detectors on the mirrored image, so that each filament can receive up to 13 masks. These extra masks only change the shape of existing detections and cannot add new ones. The masks are then drawn in order of decreasing score so that no pixel belongs to two filaments, and pieces smaller than 20 pixels are discarded.

IV. RESULTS

A. Overall results

Table II and Fig. 3 show the results of the individual models and of the system as parts are added. On their own, the four detectors reach PQ 0.42 to 0.44 and the semantic model 0.40. Combining the four calibrated detectors raises PQ to 0.450, mainly because more filaments are found (RQ increases while

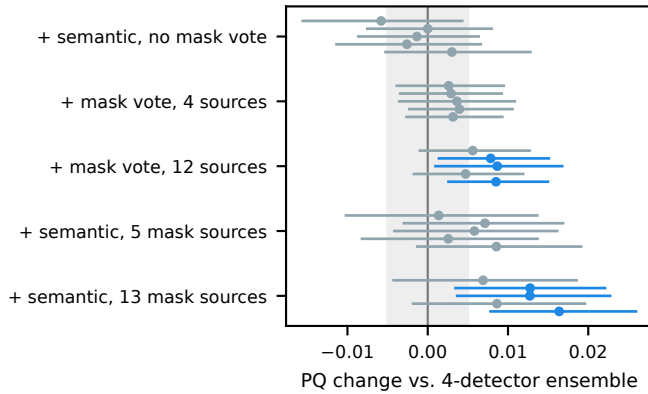


Fig. 3. Change in PQ relative to the four calibrated detectors on each of the five calibration splits (points), with paired-bootstrap 95% intervals. Blue intervals lie above zero. The gray band marks ± 0.005 , the change caused by retraining one detector with a different seed.

SQ stays the same). Adding the semantic model’s filaments directly as extra detections does not help, since they bring about as many false detections as correct ones. Voting on the masks raises PQ to 0.457 with 12 mask sources, and with the semantic model added both as a source of detections and as a mask source the full system reaches 0.461. On average it matches 600 of the 908 validation filaments, against 568 for the four detectors. The full system is better than the four detectors on all five splits, and the bootstrap interval lies above zero on three of them. With 70 validation images we consider this consistent but modest evidence.

On the public leaderboard, a single Mask R-CNN scored 0.27, adding the crop refiner raised this to 0.34, a calibrated combination of two detectors to 0.39, and the pretrained refiner to 0.40. All later versions also scored 0.40. The public score is computed on about 90 images and shown with two decimals, so differences below 0.01 cannot be seen there. Our validation scores were about 0.05 higher than the public scores for every version we submitted.

B. Ablations

Table III removes or changes one part of the system at a time, with the same five calibration splits as above. Pooling the raw scores of the four detectors without calibration lowers PQ from 0.450 to 0.428. More filaments are matched (625 against 568), but many more false detections are accepted, because a single threshold cannot suit models whose scores mean different things. Leaving out any one of the four detectors costs at most 0.002, and leaving out YOLO11l-seg costs nothing measurable, so no single detector is essential and the gain of the combination comes from their overlap.

The lower part of the table varies which sources may create detections. The semantic model helps only when it may create detections: as a source of masks alone it leaves PQ at the level of the 12-source vote (0.457), while as a source of both it raises PQ to 0.461. The opposite holds for the 512-pixel refiner and the mirrored images. When their candidates may also create detections, PQ drops from 0.457 to 0.453 with the vote and

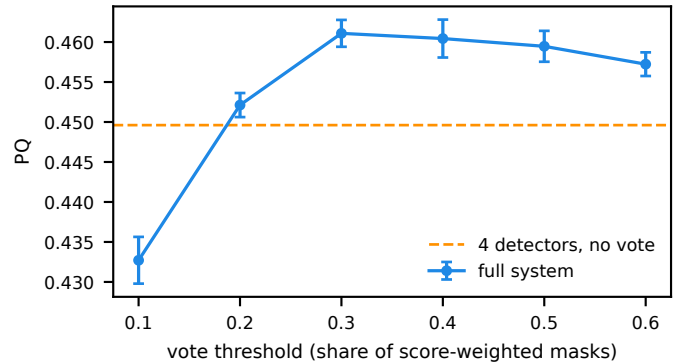


Fig. 4. PQ of the full system as a function of the vote threshold (mean and standard deviation over the five calibration splits). The dashed line is the four calibrated detectors without the vote.

TABLE III
ABLATIONS (PQ, MEAN OVER FIVE CALIBRATION SPLITS, AND CHANGE RELATIVE TO THE FOUR CALIBRATED DETECTORS).

Configuration	PQ	Δ
4 detectors, calibrated scores	.450	–
4 detectors, raw scores	.428	–.021
without Mask R-CNN	.448	–.002
without YOLO11m-seg	.448	–.002
without RT-DETR-l	.448	–.001
without YOLO11l-seg	.450	–.000
Vote, 12 mask sources, detections from 4 detectors	.457	+.007
semantic model adds masks only	.457	+.007
semantic model adds detections and masks	.461	+.012
512 px and mirrored candidates also detect	.453	+.003
the same without the vote	.448	–.002

to 0.448 without it, since they mostly add second copies of filaments that are already detected.

Fig. 4 shows the effect of the vote threshold, the share of the score-weighted masks a pixel needs in order to be kept. Values between 0.3 and 0.5 give nearly the same PQ (0.460 to 0.461), and 0.6 is only slightly lower (0.457). Below 0.3 the masks become too large and SQ falls (0.678 at 0.2 against 0.683 at 0.3), and at 0.1 PQ drops to 0.433, below the four detectors without the vote.

C. Thin structures and near misses

The organizers name fine structures such as barbs, the thin offshoots along the side of a filament, as one of the main difficulties of the task. Barbs are not labeled separately in MAGFiLO, so we approximated them by the thin parts of each annotated filament. These are the pixels removed by a morphological opening with a disk of radius 2 pixels, that is, parts narrower than about 5 pixels. They include barbs and the narrow ends of filaments and make up 1.9% of the annotated pixels. Table IV reports how many of these thin pixels, and of the remaining core pixels, are covered by the predictions.

For the four detectors, 44% of the thin pixels of detected filaments are covered, against 81% of the core pixels. Running the refiner at 512×512 instead of 256×256 hardly changes this (44.5%). The mask vote raises thin coverage to 49% and

TABLE IV

COVERAGE OF THIN PARTS (NARROWER THAN ABOUT 5 PIXELS) AND CORE PARTS OF THE ANNOTATED FILAMENTS ON ONE CALIBRATION SPLIT, BY ANY PREDICTION AND BY THE MATCHED PREDICTION OF EACH DETECTED FILAMENT (IOU ABOVE 0.5), AND THE NUMBER OF NEAR MISSES OF THE FOUR-DETECTOR SYSTEM (BEST IOU BETWEEN 0.3 AND 0.5, 96 FILAMENTS) THAT BECOME MATCHES.

Configuration	Any prediction		Matched		Near misses	
	Thin	Core	Thin	Core		
4 detectors (256px refiner)	.361	.637	.441	.813	560	–
4 detectors (512px refiner)	.364	.638	.445	.815	562	–
+ vote, 4 sources	.395	.656	.478	.832	564	11
+ vote, 12 sources	.407	.661	.491	.840	569	13
Full system	.432	.694	.502	.841	600	15
Semantic U-Net alone	.356	.569	.498	.845	462	–

core coverage to 84% without lowering SQ. So the gain does not come from simply making the masks larger. In an earlier version of the system, enlarging all masks by one pixel lowered PQ from 0.446 to 0.419. The same pattern appears when thin parts are defined with a radius of 3 pixels (coverage 0.58 for the four detectors and 0.63 for the full system). The vote by itself does not create detections. With 12 mask sources it turns 13 of the 96 near misses of the four detectors into matches and loses 4 earlier matches. The 30 additional filaments found by the full system all come from the semantic model; the four detectors had no prediction overlapping them with IoU above 0.3. The semantic model also draws the thin parts of the filaments it finds as completely as the full vote (50%), although on its own it finds far fewer filaments. Fig. 5 shows an image on which the full system separates two filaments that the four detectors merge and finds two more.

D. Organizers' overlap statistics

The challenge also evaluates the distribution of IoU and Dice over all overlapping pairs of predictions and annotations and the frequency of one-to-many and many-to-one overlaps, computed as in the organizers' evaluation notebook (Fig. 6). The mean IoU over overlapping pairs is 0.614 for the full system and 0.608 for the four detectors (mean Dice 0.744 and 0.739). The number of annotated filaments with no overlapping prediction falls from 212 to 181 of 908, while the number of predictions that overlap no annotation rises from 104 to 126. Fragmentation and merging are rare in both cases: 19 and 15 annotated filaments overlap two or more predictions, and 14 and 16 predictions overlap two or more annotations.

E. Remaining errors

On one calibration split, the four detectors miss 325 of the 908 validation filaments. For 219 of them, one of the candidates does match the filament, but its calibrated score is below the acceptance threshold (median 0.33). For 67 there is no matching candidate at all, and 57 of these have a candidate with IoU between 0.3 and 0.5, typically a mask that stops before the faint end of the filament. The remaining 39 lose out to an overlapping candidate that is wrong. Lowering the acceptance threshold does not help. Candidates with scores

TABLE V

PQ BY GONG OBSERVATORY AND BY PERIOD OF SOLAR CYCLE 24 (VALIDATION ANNOTATION SETS, CALIBRATION FITTED ON HELD-OUT HALVES). THE GROUPS ARE SMALL, SO DIFFERENCES OF A FEW HUNDREDTHS SHOULD NOT BE OVER-INTERPRETED.

Group	Sets	Filaments	Per set	4 det.	Full	Δ
Big Bear	21	141	6.7	.411	.438	+.028
Cerro Tololo	21	170	8.1	.445	.468	+.023
El Teide	22	201	9.1	.467	.472	+.005
Learmonth	14	104	7.4	.493	.500	+.007
Mauna Loa	17	124	7.3	.453	.452	−.001
Udaipur	21	168	8.0	.434	.441	+.008
2011–14 (near max.)	74	663	9.0	.447	.462	+.015
2015–18 (declining)	34	220	6.5	.465	.465	+.000
2019–22 (near min.)	8	25	3.1	.385	.404	+.019

between 0.3 and 0.5 are correct at most 32% of the time, even when nine or more sources agree on them. A detection has to be correct about 34% of the time to increase PQ.

F. Errors by filament size

Fig. 7 splits the 908 validation filaments into four groups of equal size by annotated area, again on one calibration split. The full system finds 52% of the smallest filaments (below 644 pixels) and 76% of those between 1,241 and 2,526 pixels. For the largest quarter the share drops again, to 68%. The two ends fail in different ways. Of the 110 small filaments that the full system misses, 86 have no overlapping prediction at all, so they were simply not detected. Of the 72 missed large filaments, only 20 have no overlapping prediction, and for 38 the best prediction reaches an IoU between 0.3 and 0.5. Here the prediction and the annotation clearly describe the same filament but disagree about its extent, the same kind of disagreement we find between two annotators (Section IV-H). Small predictions are also the least reliable: 56% of the predictions below 644 pixels match an annotation, against 78% of those above 2,526 pixels. The mask quality of matched filaments hardly depends on size (SQ between 0.68 and 0.70 in all four groups). The full system finds more filaments than the four detectors in every group, most clearly between 644 and 1,241 pixels (69% against 62%).

Fig. 8 shows both kinds of error on the validation image with the lowest PQ (0.21) among the images with at least ten annotated filaments. Three of the six unmatched predictions lie on annotated filaments but overlap them with an IoU between 0.30 and 0.48, so each of them counts once as a false positive and once as a missed filament. The other six missed filaments have no overlapping prediction, and several of them lie next to bright active regions.

G. Observatories and solar cycle

Table V breaks down the results by observatory and by period. PQ of the four detectors ranges from 0.41 for Big Bear to 0.49 for Learmonth, and the full system gains most at the two sites where the four detectors do worst. Images taken near the solar minimum contain few filaments (3.1 per

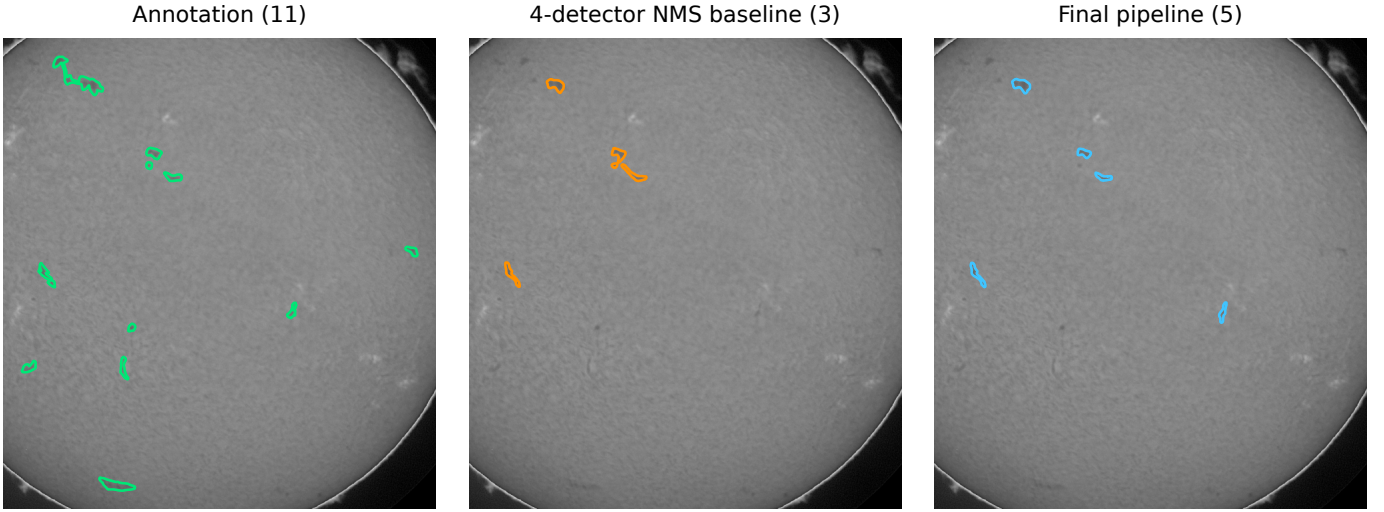


Fig. 5. A validation image on which the mask vote helps (chosen as the image with the largest gain in matched filaments, so it is not a typical case). Left: the outlines of one annotator. Middle: the four detectors merge two filaments into one object and miss several faint ones. Right: the full system separates the two filaments and finds two more, but several faint annotated filaments are still missed.

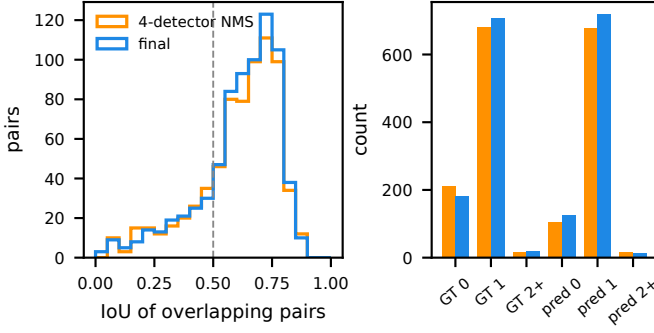


Fig. 6. The organizers’ overlap statistics on one calibration split. Left: IoU of every overlapping pair of prediction and annotation. Right: number of annotated filaments (GT) and predictions that overlap 0, 1, or 2 or more objects of the other kind. “GT 2+” corresponds to fragmentation (one-to-many) and “pred 2+” to merging (many-to-one).

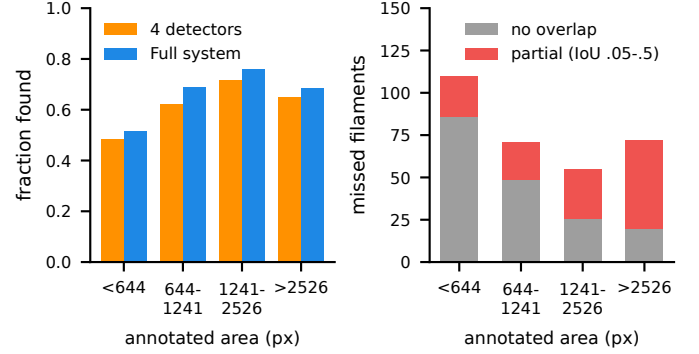


Fig. 7. Results by annotated filament area on one calibration split (quartiles of the 908 validation filaments). Left: fraction of annotated filaments found (IoU above 0.5). Right: filaments missed by the full system, split into those with no overlapping prediction and those with a partly overlapping one.

annotation set, against 9.0 near the maximum) and have the lowest scores, but there are only 8 such annotation sets.

H. Agreement between annotators

For the 296 training images annotated by two or more people, we scored each annotator’s outlines against another annotator’s with the challenge metric. The resulting PQ is 0.340 (SQ 0.634, RQ 0.536). For the 33 validation images with several annotators the value is 0.309, while our system scores 0.456 against the same annotation sets (0.447 for the four detectors). Because the models are trained on the outlines of all annotators, their output is closer to an average of the annotators than any single annotator is to another. A large part of what PQ counts as error is thus disagreement among annotators about faint structures and filament ends, which limits how high a score any method can reach on this data.

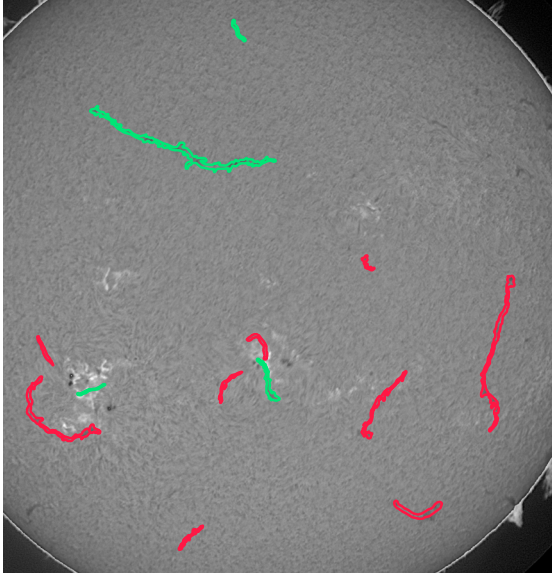
I. Alternatives that did not help

Table VI lists the main alternatives we tried and rejected. Adding further models as sources of detections, such as a second Mask R-CNN trained with another seed, proposals from mirrored images, or the semantic model without the vote, added false detections faster than correct ones. The same sources did help when they only took part in the mask vote. A gradient-boosted classifier that rescored candidates using agreement between models and shape features ranked candidates better than the calibrated scores but lowered PQ.

V. DISCUSSION

Run time. Most of the inference time goes into redrawing candidates. On average the four detectors produce 93 candidates per validation image above their minimum scores (13, 20, 41 and 19), and each is redrawn with four flipped views. On a 16 GB Kaggle GPU, Mask R-CNN together with the redrawing of its candidates takes 1.9 to 2.2 s per image, and

Annotation: 4 found, 9 missed



Full system: 4 correct, 6 false

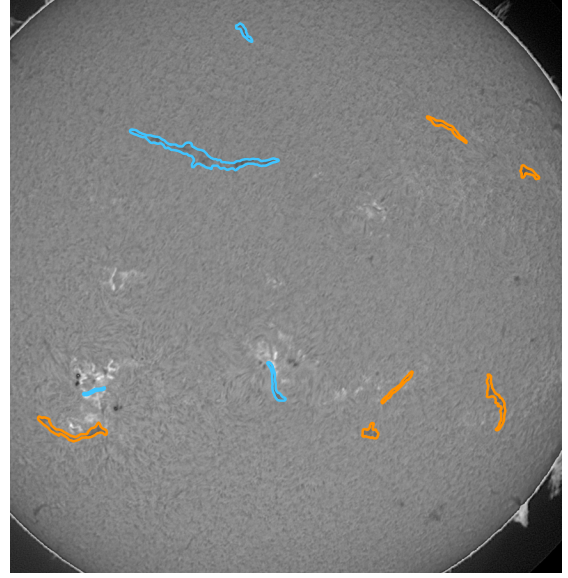


Fig. 8. A validation image on which the full system does poorly (chosen as the image with the lowest PQ among those with at least ten annotated filaments). Left: annotated filaments that the full system finds (green) and misses (red). Right: predictions that match an annotation (blue) and predictions that do not (orange).

TABLE VI

ALTERNATIVES THAT WERE TRIED AND REJECTED. CONTEXT: Y = YOLO11M-SEG ALONE; E = FOUR DETECTORS WITH THE EARLIER REFINER TRAINED FROM SCRATCH; F = FOUR DETECTORS WITH THE PRETRAINED REFINER, WITH OR WITHOUT THE MASK VOTE. PQ ON VALIDATION; RANGES OVER THE FIVE CALIBRATION SPLITS WHERE MARKED.

Alternative	Ctx.	Effect on PQ
Mask2Former (Swin-T), frozen or fine-tuned encoder	E	alone $\leq .345$; lowers E
SAM ViT-B decoder with box prompts	—	alone .316
SimSiam pretraining of the Mask R-CNN backbone on 4k unlabeled GONG images	E	no change (.448 vs .449); public .38
Self-training on pseudo-labeled test images	Y	.417 $\rightarrow$.393
CLAHE contrast enhancement	Y	.425 $\rightarrow$.412–.419
Gradient-boosted rescoring of candidates	F	–.014 to –.004 (5 splits)
Growing masks into the U-Net foreground	E / F	+.002 / –.006
Mirrored-image proposals as detections	F	–.009 to +.003 (5 splits)
Second-seed or augmented Mask R-CNN	F	–.005 to +.003 (5 splits)
Averaging two semantic models	F	–.006 to –.003 (5 splits)

the semantic model 1.2 s per image at full resolution with four flipped views. The full system redraws every candidate three times (256-pixel crops, 512-pixel crops and the mirrored image). A lighter version that uses only the five detection sources in the vote keeps about 45% of the improvement (0.005 PQ) at about a third of the redrawing cost. For processing the full GONG archive, that version may be the better choice.

Lessons for similar applications. Some of what we learned should carry over to other segmentation tasks that combine several models and have noisy labels. Scores from different models have to be calibrated before they are compared: pooling the raw scores lowered PQ by 0.021 (Table III). A model that works differently from the others helped more than further models of the same kind. The semantic model contributed all 30 filaments that no detector had found, while a second Mask R-CNN and proposals from mirrored images did not help (Table VI). Small improvements could not be judged on the public leaderboard, where every version after the first few scored 0.40, so our decisions rested on repeated held-out splits and paired resampling. Measuring the agreement between annotators early is also worthwhile. Here it showed that two annotators agree less with each other than the system agrees with them, so a large part of the remaining error cannot be removed by a better model alone.

Limitations. The acceptance and suppression thresholds and the vote parameters were chosen on the same 70 validation images on which we report results. For the vote parameters this matters little, since PQ changed by only 0.001 over a grid of 27 settings, but the other settings may be slightly optimistic. All models were trained once on one split, so our evaluation does not capture the variation from retraining. Thin structures remain the weakest point: even the full system covers only about half of the thin pixels of the filaments it finds, against 84% of the core. Neither the refiner nor the semantic model is trained to give thin structures extra weight, and barbs are also drawn inconsistently by the annotators. The 0.05 gap between our validation and public scores was stable across versions, and we have not found its cause.

Future work. For thin structures, the semantic model already draws them as completely as the full vote, so a full-resolution model trained with extra weight on thin parts is a natural next step. A topology-preserving loss (cIDice) in the crop refiner did not help in our experiments. For recall, most missed filaments do have a matching candidate, but with too low a score. Information at the scale of the whole image could help to separate these. One example is co-temporal magnetograms, which show the polarity inversion lines along which filaments lie and could be used as an additional training signal. Sequences of unlabeled GONG images could also be used for semi-supervised training with consistency between neighboring frames; self-supervised pretraining on single unlabeled frames gave no improvement in our tests.

VI. CONCLUSION

We described a system for instance segmentation of solar filaments in GONG H α images, built for the IEEE BigData Cup 2026. It combines four instance detectors and a semantic segmentation model, redraws every candidate with a pretrained crop refiner, calibrates the scores of all models, and forms each final mask by a weighted vote of the overlapping masks. On held-out validation images it reaches PQ 0.461, compared with 0.450 for the four calibrated detectors alone and 0.42 to 0.44 for the individual detectors, and it scores 0.40 on the public leaderboard without using test annotations. The mask vote improves the outlines of detected filaments, including their thin parts, and the semantic model adds filaments that the detectors miss. Thin structures and faint filament ends remain the main source of error, and the low agreement between human annotators (PQ 0.31 to 0.34) should be kept in mind when comparing results on this benchmark.

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